

Practice Exercise: Diagnosing Illness (Decision Trees)

Instructions

This practice exercise consists of four parts:

1. Test the model on new patients using the diagnostic table
2. Modify the production rules by adding a new attribute
3. Draw the updated decision tree model
4. Submit all deliverables as a ZIP file to Canvas

Test the model on new patients

Use the provided decision tree to classify the following patients. Write the diagnosis for each patient by applying the production rules shown in the decision tree diagram: [Decision tree diagram](#)

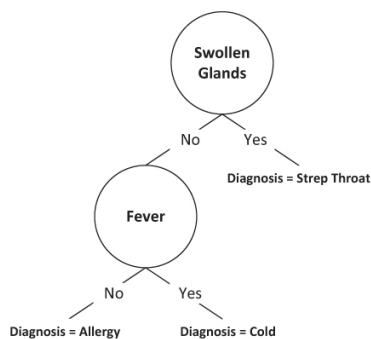


Table 1: Predictive diagnosis

Patient	Swollen Glands	Fever	Diagnosis (using the model)
14	Yes	No	
15	No	No	
16	No	Yes	
17	Yes	Yes	

For each patient, follow the branches of the tree and write the final diagnosis in the last column.

Solution

Table 2: Predictive diagnosis

Patient	Swollen Glands	Fever	Diagnosis (using the model)
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Patient	Swollen Glands	Fever	Diagnosis (using the model)
14	Yes	No	Strep throat
15	No	No	Allergy
16	No	Yes	Cold
17	Yes	Yes	Strep throat

Modify the production rules

To improve the diagnostic model, add a new attribute: **duration of symptoms**.

Use the following categories:

- **short** = fewer than 3 days
- **long** = 3 days or more

Use this attribute to refine each diagnosis as either:

- **acute** (short duration)
- **chronic** (long duration)

Original production rules:

```
IF swollen glands = YES
THEN diagnosis = strep throat

IF swollen glands = NO AND fever = YES
THEN diagnosis = cold

IF swollen glands = NO AND fever = NO
THEN diagnosis = allergy
```

Your task: Write a complete new set of production rules that incorporates the **duration** attribute.

Requirements:

- Every original rule must be expanded to include duration
- Each final rule must include all relevant conditions
- Each diagnosis must be labeled as either **acute** or **chronic**

Hint: Each original rule will split into two rules:

- one for short duration (acute)
- one for long duration (chronic)

Example (fully specified rule):

```
IF swollen glands = YES AND duration = short
THEN diagnosis = acute strep throat
```

Now write all remaining rules in the same format.

Updated production rules (6):

Solution

Updated production rules:

IF swollen glands = YES AND duration = short THEN diagnosis = acute strep throat

IF swollen glands = YES AND duration = long THEN diagnosis = chronic strep throat

IF swollen glands = NO AND fever = YES AND duration = short THEN diagnosis = acute cold

IF swollen glands = NO AND fever = YES AND duration = long THEN diagnosis = chronic cold

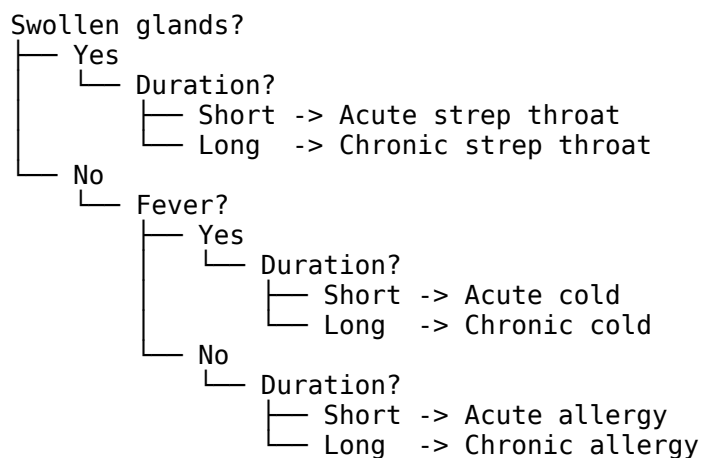
IF swollen glands = NO AND fever = NO AND duration = short THEN diagnosis = acute allergy

IF swollen glands = NO AND fever = NO AND duration = long THEN diagnosis = chronic allergy

Draw the new decision tree model

Solution

A correct updated tree should follow this logic:



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